COHOMOLOGY ON A SHEAF OF SETS COMPUTED ON THE LOCALIZED TOPOS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both assertions of the statement.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Localization of a topos at a sheaf of sets. Sites, the construction of the topos C/K attached to a sheaf of sets K and the morphism of topoi j : Sh(C/K) -> Sh(C). Supplies the morphism appearing in the statement.
- Preservation of injectives by j^{-1} and the description of cohomology from injective resolutions. Same chapter, the fact that the inverse image along a localization morphism preserves injectives, and the computation of cohomology from injective resolutions. Supplies both assertions of the statement.

Source reference. Original lemma 0791: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2101–2101. Navigation pointer invoked at proof line 2101; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04GZ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2109–2109. Navigation pointer invoked at proof line 2109; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 03F3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 608–608. Navigation pointer invoked at proof line 2113; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let K be a sheaf of sets on $\mathcal{C}$. Consider the morphism of topoi $j : Sh(\mathcal{C}/K) \to Sh(\mathcal{C})$, see Sites, Lemma 0791. Then j^{-1} preserves injectives and $H^p(K, \mathcal{F}) = H^p(\mathcal{C}/K, j^{-1}\mathcal{F})$ for any abelian sheaf $\mathcal{F}$ on $\mathcal{C}$.*

Proof. By Sites, Lemmas 04GZ and 0791 the morphism of topoi j is equivalent to a localization. Hence this follows from Lemma 03F3. □